Biochemistry Cume

January 16, 1999

Cascades:

Blood clotting, visual excitation, complement fixation and glycogen metabolism are all mediated by cascades of enzyme-catalyzed reactions.

a. (10 points) What are several essential common features of cascades?

b. (20 points) Describe and explain important biological advantages that are conferred by reaction cascades.

c. (20 points) Describe any two of the types of covalent protein modification that are used in cascades. For each type, identify the chemical reaction and the site where the protein is modified, and state whether the process is reversible or irreversible, and why.

Coenzymes.
Choose any five the twelve coenzymes from the list below:

a. (2 points each) State an example of a type of reaction or enzyme that requires the coenzyme.

b. (4 points each) Show or describe the critical chemical functional group on the coenzyme.

c. (4 points each) Indicate as much of the reaction mechanism associated with this functional group as you can.

Choose 5 of 12:

ATP	Lipoic acid
NADH	Cobalamine (B_{12})
Coenzyme A	FAD
Biotin	Folate
Thiamine	S-adenosylmethionine
Pyridoxal (vitamin B_6)	Retinal (vitamin A)